

Spectral Distributions of Random Matrix Ensembles

Random Matrix Theory Group (SMALL 2024 REU)

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Classical Random Matrix Ensembles

Consider an $N \times N$ real symmetric matrix A :

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1N} \\ a_{12} & a_{22} & a_{23} & \cdots & a_{2N} \\ a_{13} & a_{23} & a_{33} & \cdots & a_{3N} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{1N} & a_{2N} & a_{3N} & \cdots & a_{NN} \end{pmatrix}, \quad \text{where } A = A^\top, a_{ij} = a_{ji}$$

Fix a probability distribution $p(a_{ij})$ for the entries a_{ij} .

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Fix a probability distribution $p(a_{ij})$ for the entries a_{ij} .

Probability Measure

$$\text{Prob}(A) = \prod_{1 \leq i \leq j \leq N} p(a_{ij})$$

This means that for any intervals $[\alpha_{ij}, \beta_{ij}]$,

$$\text{Prob}(A : a_{ij} \in [\alpha_{ij}, \beta_{ij}]) = \prod_{1 \leq i \leq j \leq N} \int_{\alpha_{ij}}^{\beta_{ij}} p(x_{ij}) dx_{ij}$$

Goal: Understand the eigenvalues of A .

Historical Background and Importance

- **1920s: John Wishart** introduced random matrices in multivariate statistics [**Wishart**].
- **1950s: Eugene Wigner** developed the **semi-circle law** while studying nuclear physics [**Wigner1**, **Wigner2**].

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Significance

- RMT provides insights into complex systems in physics, number theory, and data science.
- Understanding eigenvalue distributions aids in modeling phenomena like energy levels in heavy nuclei and zeros of the Riemann zeta function.

Current Applications of Random Matrix Theory

- **Physics:**

- Quantum chaos and energy level statistics.
- Modeling disordered systems.

- **Number Theory:**

- Connections to zeros of the Riemann zeta function.
- Insights into prime number distributions.

- **Data Science and Machine Learning:**

- Principal Component Analysis (PCA) in high-dimensional data.
- Neural network weight initialization.

- **Wireless Communications:**

- Analyzing multiple-input multiple-output (MIMO) systems.

Trace of Matrix Powers

Definition (Trace of a Matrix)

The **trace** of an $N \times N$ matrix A , denoted by $\text{Tr}(A)$, is the sum of its diagonal elements:

$$\text{Tr}(A) = a_{11} + \cdots + a_{NN}.$$

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Theorem (Trace of Matrix Powers)

Let A be an $N \times N$ matrix. Then,

$$\text{Tr}(A^k) = \sum_{i_1=1}^N \sum_{i_2=1}^N \cdots \sum_{i_k=1}^N a_{i_1 i_2} a_{i_2 i_3} \cdots a_{i_{k-1} i_k} a_{i_k i_1}.$$

Eigenvalue Trace Formula

Theorem (Eigenvalue Trace Formula)

For any non-negative integer k , if A is an $N \times N$ matrix with eigenvalues $\lambda_i(A)$, then

$$\text{Tr}(A^k) = \sum_{i=1}^N \lambda_i(A)^k.$$

Proof of the Eigenvalue Trace Formula

Proof of Eigenvalue Trace Formula.

$k = 1$ follows from the definition of trace:

$$\mathrm{Tr}(A) = \sum_{i=1}^N \lambda_i(A).$$

For higher powers k , consider that any matrix A can be diagonalized via a similarity transformation:

$$U^{-1}AU = T,$$

where U is unitary and T is upper triangular with eigenvalues $\lambda_i(A)$ on diagonal.

$$\implies U^{-1}A^kU = T^k.$$



Proof of the Eigenvalue Trace Formula

Proof of Eigenvalue Trace Formula (contd.)

Since the trace is invariant under similarity transformations:

$$\mathrm{Tr}(A^k) = \mathrm{Tr}(T^k).$$

The matrix T^k remains upper triangular with diagonal entries $\lambda_i(A)^k$, thus:

$$\mathrm{Tr}(T^k) = \sum_{i=1}^N \lambda_i(A)^k.$$



Eigenvalue Probability Distribution

Definition (Eigenvalue Probability Distribution)

For each $N \times N$ matrix A , we associate a probability measure $\mu_{A,N}(x)$ defined as:

$$\mu_{A,N}(x) dx = \frac{1}{N} \sum_{i=1}^N \delta \left(x - \frac{\lambda_i(A)}{\sqrt{N}} \right) dx. \quad (1.26)$$

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- At each normalized eigenvalue $\frac{\lambda_i(A)}{\sqrt{N}}$, a mass of weight $\frac{1}{N}$ is placed.
- Since there are N masses, $\mu_{A,N}(x)$ forms a probability distribution.
- For a probability distribution $p(x)$, the integral $\int_a^b p(x) dx$ represents the probability of observing a value in the interval $[a, b]$.

$$\int_a^b \mu_{A,N}(x) dx = \frac{\# \left\{ i : \frac{\lambda_i(A)}{\sqrt{N}} \in [a, b] \right\}}{N}. \quad (1)$$

Moments of the Eigenvalue Probability Distribution

Definition (Moment of Eigenvalue Probability Distribution)

Let $E[x^k]_A$ denote the k -th moment of the eigenvalue probability distribution $\mu_{A,N}(x)$:

$$E[x^k]_A = \int_{-\infty}^{\infty} x^k \mu_{A,N}(x) dx.$$

We often denote this moment as $M_{N,k}(A)$.

Remark

Calculating the moments $M_{N,k}(A)$ allows us to analyze the statistical properties of the eigenvalue distribution $\mu_{A,N}(x)$. These moments are instrumental in understanding the convergence and behavior of spectral measures in the asymptotic limit as $N \rightarrow \infty$.

Moments of the distribution

Lemma (Computing Moments by Trace)

Let $M_{N,k}$ be the k^{th} moment of the spectral distribution $\nu_{A_N,N}$. The moment can be computed by the trace of A_N . i.e.

$$M_{N,k}(A_N) = \frac{\lambda_1^k + \cdots + \lambda_N^k}{N^{\frac{k}{2}+1}} = \frac{\text{Tr}(A_N^k)}{N^{\frac{k}{2}+1}}.$$

Remark

The power of N on the numerator depends on the normalization adopted for the definition of spectral density.

Proof of the Moment Formula

Proof by Miller and Takloo-Bighash (2010).

As $\text{Tr}(A^k) = \sum_i \lambda_i(A)^k$, we have

$$\begin{aligned} M_{N,k}(A) &= \mathbb{E}[x^k]_A = \int x^k \mu_{A,N}(x) dx \\ &= \frac{1}{N} \sum_{i=1}^N \int_{\mathbb{R}} x^k \delta\left(x - \frac{\lambda_i(A)}{\sqrt{N}}\right) dx \\ &= \frac{1}{N} \sum_{i=1}^N \left(\frac{\lambda_i(A)^k}{(\sqrt{N})^k} \right) \\ &= \frac{\text{Tr}(A^k)}{N^{\frac{k}{2}+1}}. \end{aligned}$$



Matrix to RMT Ensembles

Corollary

For a random matrix ensemble X_N , the moment of the spectral density can be computed by the expected trace:

$$M_{N,k}(X_N) = \frac{\mathbb{E}[\text{Tr}(X_N^k)]}{N^{\frac{k}{2}+1}}.$$

Matrix to RMT Ensembles

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Definition (Limiting Spectral Density of a RMT Ensemble)

The LSD is defined as the spectral density of the RMT ensemble as N approaches infinity. Also, we define the moment of the LSM as follows.

$$\nu_X(x) = \lim_{N \rightarrow \infty} \nu_{X_N, N}(x).$$

LSD's are well-defined for most matrix ensembles.

Wigner's Semi-Circle Law

Question

Returning back to our original question: how are eigenvalues in a real symmetric random ensemble distributed?

Wigner's Semi-Circle Law

Question

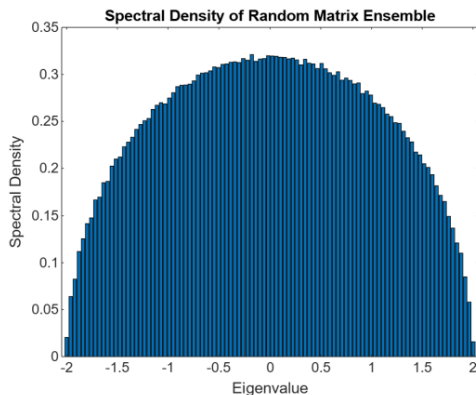
Returning back to our original question: how are eigenvalues in a real symmetric random ensemble distributed?

Theorem (Wigner's Semi-Circle Law)

Consider the ensemble of $N \times N$ real symmetric matrices with entries independently chosen from a fixed probability density $p(x)$ with mean 0, variance 1, and finite higher moments. As $N \rightarrow \infty$, for almost all A , the spectral measure $\nu_{A,N}(x)$ converges to the semi-circle density:

$$\rho(x) = \frac{1}{2\pi} \sqrt{4 - x^2}, \quad \text{for } |x| \leq 2.$$

Experimental Spectral Density: $N = 1000$



$$2/2\pi \approx 0.318$$

Random Matrix Ensembles: the GOE

Definition (Gaussian Orthogonal Ensemble)

The GOE X_N is constructed by assigning a random variable a_{ij} to each entry of a square matrix by the following rules:

$$\begin{cases} a_{ij} = a_{ji} \sim \mathcal{N}(0, 1) & i \neq j \\ a_{ii} \sim \mathcal{N}(0, 2) \end{cases}.$$

$$X_N = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1N} \\ a_{21} & a_{22} & \cdots & a_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ a_{N1} & a_{N2} & \cdots & a_{NN} \end{pmatrix}.$$

For any instance A_N^1 of X_N , $A_N = A_N^\top$, so eigenvalues of A_N are purely real.

¹ A_N is a fixed matrix w/o randomness

Deriving Spectral distributions

Definition (Spectral Measure of Eigenvalues)

$$\nu_{A_N, N}(x) = \frac{1}{N} \sum_{i=1}^N \delta \left(x - \frac{\lambda_i}{\sqrt{N}} \right),$$

where $\{\lambda_i\}_{i=1}^N$ are the eigenvalues of X_N . Size of eigenvalues are typically $\Theta(\sqrt{N})$.

LSD of GOE: Semicircular Law

Theorem (Semicircular Law)

Let $\{X_N\}_{N=1}^\infty$ be a sequence of $N \times N$ GOE random matrices with spectral measure $\{\nu_{X_N, N}\}_{N=1}^\infty$. Then, $\{\nu_{X_N, N}\}_{N=1}^\infty$ converges weakly almost surely to semicircle distribution

$$\lim_{N \rightarrow \infty} \nu_{X_N, N} = \sigma,$$

where $\sigma := \frac{1}{2\pi} \sqrt{4 - t^2}$, in the sense that

$$\lim_{N \rightarrow \infty} \mathbb{P}(|M_{N,k}(X_N) - M_k(\sigma)| > \epsilon) = 0.$$

A Useful Tool

Lemma (Wick's Formula)

Let $(x_1, \dots, x_n)$ be a real Gaussian random vector, and $\mathcal{P}_2(k)$ be the set of all pairings of $[k]$. Then

$$\mathbb{E}(x_{i_1} \cdots x_{i_k}) = \sum_{\pi \in \mathcal{P}_2(k)} \mathbb{E}_{\pi}(x_{i_1}, \dots, x_{i_k}) \quad \text{for any } i_1, \dots, i_k \in [n],$$

where $\mathbb{E}_{\pi}$ is the paired expectation, i.e.

$$\mathbb{E}_{(12)(34)}[x_1 x_2 x_3 x_4] = \mathbb{E}[x_1 x_2] \mathbb{E}[x_3 x_4].$$

Proof Sketch of the semicircle law

Lemma (Moments of GOE)

$$M_k(\sigma) = \frac{1}{2\pi} \int_{-2}^2 t^k \sqrt{4-t^2} dt = \begin{cases} C_{k/2} & (k = 0 \pmod{2}) \\ 0 & (k = 1 \pmod{2}) \end{cases},$$

where C_k is the k^{th} Catalan number.

Proof of the Semicircle Law

Proof.

The trace can be expanded into product of the entries. Let $(i_n)_{1 \leq n \leq k}$ be a sequence of integers where $1 \leq i_n \leq N$. Then,

$$\mathbb{E} [\text{tr}(X_N^k)] = \mathbb{E} \left[\sum_{(i_n)} \prod_{1 \leq n \leq k} x_{i_n, n+1} \right] = \mathbb{E} \left[\sum_{(i_n)} x_{i_1, i_2} x_{i_2, i_3} \cdots x_{i_n, i_{n+1}} \right]$$

where $i_{n+1} = i_1$ for convenience. Upon inspection, odd moments vanish by a parity argument. We assume k to be even and write $2k$ from now on.

Invoking Wick's formula, write

$$= \sum_{\pi \in \mathcal{P}_2(2k)} \mathbb{E}_{\pi} \left[\prod_{(i_n)} x_{i_n, i_{n+1}} \right]$$

Proof of the Semicircle Law

Proof (continued).

The paired expectation will not vanish iff all the matchings relate two identical entries. For example, consider the paired expectation of

$$x_{14}x_{43}x_{34}x_{41}$$

then the only $\pi \in \mathcal{P}(4)$ where $\mathbb{E}_\pi[\prod x]$ is nonvanishing is

$$\pi = (14)(23)$$

Using the language of group theory, the paired expectation is 1 if and only if

$$\gamma_k \circ \pi(i_n) = i_n \quad \forall 1 \leq n \leq k$$

where $\gamma_k = (123 \cdots k)$, or $\gamma(n) = n + 1 \pmod n$

Proof of the Semicircle Law

Proof (continued).

Thus, to compute the expected trace, we can count the number of sequences that are invariant under the permutation $\gamma \circ \pi$.

$$\begin{aligned}\mathbb{E}(\text{Tr}(X^{2k})) &= \sum_{i_1, \dots, i_{2k}=1}^N |\{\pi \in \mathcal{P}_2(2k) \mid i \text{ is constant on the orbits of } \gamma_{2k}\pi\}| \\ &= \sum_{\pi \in \mathcal{P}_2(2k)} |\{i : [2k] \rightarrow [N] \mid i \text{ is constant on the orbits of } \gamma_{2k}\pi\}| \\ &= \sum_{\pi \in \mathcal{P}_2(2k)} N^{\#(\gamma_{2k}\pi)}\end{aligned}$$



Proof of the Semicircle Law

Proof (continued).

We invoke Biane's result. The length of the permutations obey the triangle inequality. We convert the length of permutations into the number of orbits. Also, $\pi = \pi^{-1}$.

$$\begin{aligned} |\gamma_{2k}| &\leq |\pi| + |\gamma_{2k}\pi^{-1}| \\ 2k - 1 &\leq k + 2k - \#(\gamma_{2k}\pi) \\ (\gamma_{2k}\pi) &\leq k + 1 \end{aligned}$$

The equality holds if and only if π is non-crossing.
Thus,

$$\mu_{2k}(X) = \frac{\mathbb{E}[\text{Tr}(X^{2k})]}{N^{k+1}} = |NC(2k)| = C_k$$



Combining two RMT Ensembles

Question

Is there a natural way to combine two random matrix ensembles such that

- ① *All the eigenvalues are real;*
- ② *The combination is symmetric.*

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Definition

Consider the Anticommutator product, namely

$$\{A, B\} := AB + BA.$$

Anticommutator of two RMT Ensembles

- ❶ GOE;
- ❷ Palindromic Toeplitz;
- ❸ k -checkerboard.

Spectral Density of the Anticommutator

Definition (Spectral Density of the Anticommutator)

$$\mu_N(x) := \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{\lambda \in \Lambda} \delta \left(x - \frac{\lambda}{N} \right).$$

Theorem (Moments of Spectral Density)

$$M_{N,k}(X_N Z_N + Z_N X_N) = \mathbb{E} \left[\text{Tr} \left(\left[\frac{1}{N} (X_N Z_N + Z_N X_N) \right]^k \right) \right].$$

Definition of PTE

Definition (Palindromic Toeplitz)

An $N \times N$ real symmetric palindromic Toeplitz matrix (where N is assumed to be even for simplicity) is a matrix A_N whose entries are parametrized by $b_0, b_1, \dots, b_{N/2-1}$, where the b_i 's are i.i.d. random variables with mean 0 and variance 1:

$$a_{ij} = \begin{cases} b_{|i-j|}, & \text{if } 0 \leq |i-j| \leq \frac{N}{2} - 1 \\ b_{N-1-|i-j|}, & \text{if } \frac{N}{2} \leq |i-j| \leq N-1. \end{cases}$$

This matrix is in the form

$$\begin{pmatrix} b_0 & b_1 & \cdots & b_1 & b_0 \\ b_1 & b_0 & \cdots & b_2 & b_1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ b_1 & b_2 & \cdots & b_0 & b_1 \\ b_0 & b_1 & \cdots & b_1 & b_0 \end{pmatrix}.$$

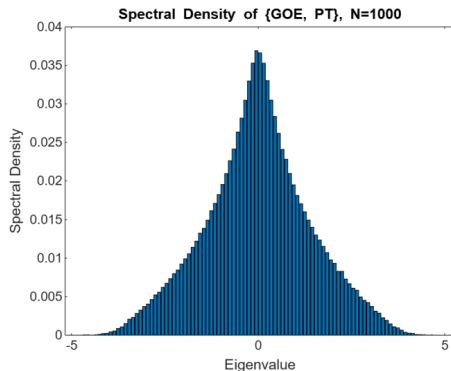
Remark (Kologlu, Kopp, Miller 2011)

*The structure of the PTE creates Diophantine Obstructions which make certain terms of the expected trace vanish in the $N \rightarrow \infty$. In particular, terms that have a **Crossing Pairing** vanish.*

Moments of $\{GOE, PT\}$

Question

Can we compute the moments of the Spectral distribution of the anticommutator of two ensembles GOE, PT ?



2th moment: 2.005185
3th moment: -0.000116
4th moment: 12.220592
5th moment: -0.059222
6th moment: 110.056541
7th moment: 2.953869
8th moment: 1177.779577

Figure: Moments of $\{GOE, PTE\}$

Computing Normalized Spectral Density

Remark

Foiling trace expansion and invoking Wick's formula, we verify that GOE's have crossings do not contribute to the moments as $N \rightarrow \infty$.

Example

$$\begin{aligned} N^3 \mu(2) &= \mathbb{E}[\text{Tr}(XZXZ)] + \mathbb{E}[\text{Tr}(XZZX)] + \mathbb{E}[\text{Tr}(ZXXZ)] + \mathbb{E}[\text{Tr}(ZXZX)] \\ &= 2 (\mathbb{E}[\text{Tr}(XXZZ)] + \mathbb{E}[\text{Tr}(XZXZ)]). \end{aligned}$$

The following computations give motivation for the following definitions.

Special Words

Definition (Special Words)

A special word of length $2k$ is composed of k blocks of $\{XX, ZX, XZ\}$. The characteristic of a special word w , $\chi(w)$, is the number XX blocks.

Example

When $k = 3$,

$XX \ ZX \ XZ$

has length 6, characteristic 1.

Set of Special Words

Definition (Set of Special Words)

$H_{n,k}$ is the set of special words of length $2n$, characteristic k .

Example

For $n = 2$ and $k = 1$:

$$H_{2,1} = \{XXZX, XX XZ, ZX XX, XZ XX\}.$$

Valid Pairings

Definition (Valid Pairings)

A pairing is valid if each paired letter are the same.

Example

For the word $XX ZZ$, a valid pairing is

$$(12)(34)$$

and an invalid pairing is

$$(13)(24).$$

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Remark (Motivation for Validness)

Assuming x, z are independent r.v. with mean zero, then

$$\mathbb{E}[xz] = 0.$$

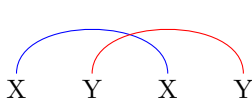
Non-Crossing Pairings

Definition (Non-Crossing Pairings)

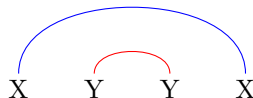
A non-crossing pairing is a valid pairing where for any two pairs $\{i, k\}$ and $\{j, l\}$, it is not the case that $i < j < k < l$.

Example

For the word $XXZZ$, the pairing $(12)(34)$ is non-crossing, while the pairing $(13)(24)$ is crossing because $1 < 2 < 3 < 4$.



Crossing



Non-Crossing

The crossing pairs vanish due to Diophantine Obstructions.

Pairing Number

Definition (Pairing Number)

$\nu_{n,k}$ is the number of valid, non-crossing pairings for all words in $H_{n,k}$, i.e.

$$\nu_{n,k} = \sum_{w \in H_{n,k}} \varphi(w),$$

where $\varphi(w)$ counts valid, non-crossing pairings of w .

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Definition (Auxiliary Sequence)

Define

$$\sigma_{n,s,k} := \sum_{w \in H_{n,s,k}} \varphi(w).$$

$H_{n,s,k}$ is the set of all words with n total blocks, at least s blocks of XX in the beginning, k blocks of XZ, ZX .

Moment Computation as a Combinatorial Problem

Theorem (Moments of $\{PTE, GOE\}$)

The n^{th} moment of the LSD of the anticommutator ensemble of $\{GOE, PTE\}$ can be computed by the pairing number, i.e.

$$M_n = \nu_{n,n} = \sigma_{n,s=0,k=-n}.$$

Initial Conditions for $\sigma_{n,s,k}$

Theorem (Initial Conditions for $\sigma_{n,s,k}$)

For $n, s, k \in \mathbb{Z}_{\text{pos}}$,

- $\sigma_{n,s,k} = 0$ if $s + k > n$,
- $\sigma_{n,s,2k+1} = 0$,
- $\sigma_{n,s,-k} = 0$,
- $\sigma_{n,s,0} = (2n - 1)!!$.

Proof.

Conditions $s + k > n$ and $k < 0$ will not generate a valid word. k odd does not generate valid pairings.

If $k = 0$, then the word is comprised solely of X 's, which reduces to the GOE case. □

Theorem 3: Recurrence Relation for $\sigma_{n,s,2k}$

Theorem (Recurrence Relation for $\sigma_{n,s,2k}$)

The recurrence relation for $\sigma_{n,s,2k}$ is given by:

$$\begin{aligned}
\sigma_{n,s,2k} = & \sum_{p=s+1}^n \sum_{q=p+1}^n \sum_{r=0}^{2k} [\sigma_{n-q+p,p,r} \cdot \sigma_{q-p-1,0,2k-2-r}] \\
& + \sum_{p=s+1}^n \sum_{q=p+1}^n \sum_{r=0}^{2k} [\sigma_{n-q+p-1,p-1,r} \cdot \sigma_{q-p,1,2k-2-r}].
\end{aligned}$$

Recurrence Relation: Demo

Proposition

The pairing number of the word is given by the sum of the product of pairing numbers generated by the Y -slicings.

$$\varphi(W) = \sum_{\substack{W_1, V_1 \\ \text{Type 1}}} \varphi(W_1)\varphi(V_1) + \sum_{\substack{W_2, V_2 \\ \text{Type 2}}} \varphi(W_2)p(V_2).$$

Recurrence Relation: Demo

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Corollary (Informal justification of the recurrence relation)

$$\sigma_{n,s,2k} = \sum_{p=s+1}^n \sum_{q=p+1}^n \sum_{r=0}^{2k} \left[\sigma_{n-q+p,p,r} \cdot \sigma_{q-p-1,0,2k-2-r} + \sigma_{n-q+p-1,p-1,r} \cdot \sigma_{q-p,1,2k-2-r} \right].$$

Numerical computations of theoretical moments

Table of $\sigma_{n,k}$ for $N = 15$:

k:	0	1	2	3	4	5
n						
0:	1					
1:	0	1				
2:	2	0	3			
3:	0	12	0	15		
4:	12	0	84	0	105	
5:	0	160	0	720	0	945
6:	104	0	1908	0	7470	010395

Figure: Numerical values of $\sigma_{n,s=0,k}$

Thanks

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